F1 Score

A Complete Guide to the Harmonic Mean of Precision & Recall

The F1 Score is one of the most widely used metrics in machine learning and statistics for evaluating classification models. It elegantly combines two competing concerns — precision and recall — into a single, interpretable number.

Metric Type	Classification Evaluation
Range	0.0 → 1.0 (higher is better)
Formula	2 × (Precision × Recall) / (Precision + Recall)
Best For	Imbalanced datasets
Variants	Macro, Micro, Weighted, F-beta

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1. Background — Precision, Recall & the Confusion Matrix

Before defining F1, we need two building blocks: **Precision** and **Recall**. Both are derived from the **Confusion Matrix** — a 2×2 table that tallies every prediction a classifier makes on a binary task.

	Predicted Positive	**Predicted Negative**
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Figure 1 — The Confusion Matrix

Precision — Of all the instances the model labeled positive, what fraction truly were positive?

$$\text{Precision} = TP / (TP + FP)$$

High precision means the model rarely raises false alarms. A spam filter with high precision will almost never flag a legitimate email as spam.

Recall (Sensitivity) — Of all actual positives, what fraction did the model find?

$$\text{Recall} = TP / (TP + FN)$$

High recall means the model misses very few real positives. A cancer-screening test with high recall catches almost all patients who are actually ill.

The Precision–Recall Trade-off
Increasing the classification threshold raises precision (fewer false alarms) but lowers recall (more missed positives). Decreasing the threshold does the opposite. No single threshold maximises both simultaneously — hence the need for a combined metric like F1.

2. Deriving the F1 Score

The F1 Score is the **harmonic mean** of Precision and Recall. Unlike the arithmetic mean, the harmonic mean penalises extreme imbalances between the two values.

Core Formula

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Equivalently, using TP, FP, FN directly:

$$\text{F1} = 2 \cdot \text{TP} / (2 \cdot \text{TP} + \text{FP} + \text{FN})$$

Why the Harmonic Mean?

If either Precision or Recall is zero, the F1 Score is also zero — even if the other metric is perfect. The arithmetic mean would give 0.5 in that scenario, which would be misleadingly optimistic. Consider the comparison below:

Scenario	Precision	Recall	Arith. Mean	F1 (Harm. Mean)
Balanced model	0.80	0.80	0.80	0.80
One metric poor	0.90	0.10	0.50	0.18
One metric zero	1.00	0.00	0.50	0.00
Both high	0.92	0.88	0.90	0.90

Table 1 — Harmonic vs Arithmetic Mean comparison

Step-by-Step Worked Example

A fraud-detection classifier on a test set of 10 000 transactions produces: TP = 180, FP = 20, FN = 60, TN = 9 740.

- Precision = 180 / (180 + 20) = 180 / 200 = **0.90**
- Recall = 180 / (180 + 60) = 180 / 240 = **0.75**
- F1 = 2 × 0.90 × 0.75 / (0.90 + 0.75) = 1.35 / 1.65 ≈ **0.818**

The F1 of 0.818 correctly reflects that recall is dragging down performance — the model misses 60 real fraudulent transactions out of 240.

3. Interpreting F1 | 4. Variants

Interpreting the Score

F1 Range	Interpretation
0.90 – 1.00	Excellent — production-ready for most tasks
0.80 – 0.89	Good — suitable for many applications
0.70 – 0.79	Moderate — consider further tuning
0.50 – 0.69	Poor — significant room for improvement
Below 0.50	Failing — re-examine data, features, or model choice

Table 2 — F1 Score Interpretation Guide

4. Variants of F1

In multi-class settings there is no single "positive" class. Averaging strategies determine how per-class F1 values are combined.

Variant	How Computed	Best Used When
Macro F1	Unweighted mean of per-class F1	All classes equally important
Micro F1	Global TP, FP, FN pooled across classes	Overall performance matters most
Weighted F1	Class-frequency-weighted mean of F1	Class imbalance is present
F-beta (Fβ)	$(1+\beta^2)\times P\times R / (\beta^2\times P + R)$	Recall (β>1) or Precision (β<1) is prioritised

Table 3 — F1 Averaging Strategies

The F-beta Score

$$F_\beta = (1 + \beta^2) \times (\text{Precision} \times \text{Recall}) / (\beta^2 \times \text{Precision} + \text{Recall})$$

β controls the weight given to recall relative to precision. **F2** (β=2) weights recall twice as heavily — ideal for medical screening where missing a disease (false negative) is costlier than a false alarm. **F0.5** (β=0.5) weights precision more — useful in recommendation systems where irrelevant suggestions damage user trust.

5. Real-World Applications

Medical Diagnosis	Disease classifiers (cancer, COVID-19) operate on severely imbalanced datasets where true positives are rare. Accuracy alone is misleading — a model that predicts "healthy" for every patient achieves >99% accuracy on a dataset with 1% disease prevalence. F1 (or F2, favouring recall) forces the model to detect real cases.
Spam Detection	Email filters need high precision (don't block legitimate mail) but also reasonable recall (catch actual spam). F1 provides a single number to compare filter versions; teams often track the full Precision–Recall curve and choose a threshold that maximises F1.
Named Entity Recognition (NLP)	NER models tag entities such as person names, organisations, and locations in text. Performance is measured token-by-token: every predicted entity span must match a ground-truth span exactly. Micro-F1 is the standard benchmark metric on datasets like CoNLL-2003.
Information Retrieval	Search engines return a ranked list of documents. At a given cutoff k, precision@k and recall@k can be combined into F1@k. This guides index tuning and ranking model selection, ensuring relevant documents are surfaced without excessive noise.

Practical Python Code

```
from sklearn.metrics import f1_score, classification_report

y_true = [0, 1, 1, 0, 1, 1, 0, 0, 1, 0]
y_pred = [0, 1, 0, 0, 1, 1, 1, 0, 1, 0]

# Binary F1
f1 = f1_score(y_true, y_pred)
print(f"F1 Score: {f1:.4f}")

# Full report (multi-class)
print(classification_report(y_true, y_pred))
```

6. Limitations & When to Use Other Metrics

F1 is powerful, but no metric is universally best. Understanding its weaknesses helps you choose the right evaluation strategy.

■ Does Not Account for True Negatives

F1 ignores TN entirely. In tasks where correct negative predictions matter (e.g., patient safety monitoring), Matthews Correlation Coefficient (MCC) or AUC-ROC may be more informative.

■ Threshold Sensitivity

F1 is computed at a fixed decision threshold (default 0.5). The Area Under the Precision–Recall Curve (AUPRC) provides a threshold-agnostic view and is preferred when comparing models across operating points.

■ Treats FP and FN Equally

Standard F1 gives equal weight to false positives and false negatives. When one error type is far costlier (e.g., missing cancer vs. unnecessary biopsy), the F-beta score should be used with a β tuned to the cost ratio.

■ Multi-class Ambiguity

Macro, micro, and weighted F1 can tell very different stories. Always report which averaging method you use and complement with per-class F1 when class-level performance matters.

■ Class Imbalance Nuance

Even weighted F1 can be inflated by a dominant majority class. For extreme imbalance (99:1 ratio), AUPRC or balanced accuracy may be better choices.

Metric Selection Guide

Situation	Recommended Metric
Balanced classes, overall performance	Accuracy or Macro F1
Imbalanced classes	Weighted F1 or AUPRC
False negatives much costlier (e.g., disease)	Recall, F2 Score
False positives much costlier (e.g., spam)	Precision, F0.5 Score
Binary, threshold-independent comparison	AUC-ROC or AUPRC
TN matters (safety systems)	MCC
Multi-class, class imbalance	Per-class F1 + Weighted F1

Key Takeaways

- F1 = harmonic mean of Precision and Recall; range 0–1, higher is better.
- Use it whenever class imbalance makes accuracy misleading.
- Choose macro/micro/weighted averaging intentionally in multi-class tasks.
- Adjust β in F-beta when false negatives and false positives carry unequal costs.
- Pair F1 with AUPRC or MCC for a full picture of model performance.